

Water Quality Compliance Classification

Predictive Analysis Using Machine Learning

5CS037 - Concepts and Technologies of AI

Final Portfolio Project 2026

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Abstract

This report presents a comprehensive classification analysis of water quality compliance using machine learning techniques. The dataset comprises 81,026 water quality measurements from England (January 2025), containing 13 attributes related to water sampling and quality assessment. The project directly supports UN Sustainable Development Goal 6 (Clean Water and Sanitation) by developing automated compliance prediction models. Three classification approaches were implemented: a Neural Network achieving 96.7% accuracy, and two classical machine learning models—Logistic Regression (97.2% accuracy) and K-Nearest Neighbors (99.1% accuracy). Through systematic hyperparameter optimization using GridSearchCV and feature selection via Recursive Feature Elimination (RFE), the models were refined to use only 5 critical features while maintaining high performance. Cross-validation confirmed robust generalization across all models. The K-Nearest Neighbors model emerged as the best performer with 99.1% accuracy, 99.1% precision, 99.1% recall, and 99.1% F1-score, demonstrating exceptional capability for real-world deployment in automated water quality monitoring systems.

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1. Introduction

1.1 Problem Statement

Water quality monitoring is critical for public health and environmental protection. This project addresses the challenge of automating compliance classification of water samples based on multiple quality parameters. The goal is to predict whether a water sample meets regulatory compliance standards (`sample.isComplianceSample = True`) based on physical, chemical, and contextual features. Manual inspection of water quality data is time-consuming and resource-intensive; automated classification can significantly reduce inspection costs while improving detection accuracy.

1.2 Dataset

The England Water Quality Sampling Dataset (2025) contains 81,026 records with 13 attributes including sampling location, measurement results, determinand types, material types, and compliance status. Data was collected from environmental agencies monitoring sewage effluent and surface water quality across England during January 2025. The dataset includes both compliance audit samples and environmental monitoring samples, providing a balanced view of regulatory and ecological water quality assessment.

1.3 UN SDG Alignment

This project directly supports UN Sustainable Development Goal 6: Clean Water and Sanitation, specifically Target 6.3 which aims to improve water quality by reducing pollution and increasing treatment. By developing automated compliance monitoring tools, this work enables faster identification of non-compliant samples, supports regulatory enforcement, and contributes to environmental protection and public health safety.

1.4 Objectives

The primary objectives of this analysis are: (1) Build and evaluate a Neural Network classifier for compliance prediction; (2) Develop two classical machine learning models (Logistic Regression and K-Nearest Neighbors); (3) Optimize model performance through hyperparameter tuning; (4) Identify the most important features using feature selection techniques; (5) Compare final models and recommend the best approach for deployment.

2. Methodology

2.1 Data Preprocessing

Data preprocessing involved several critical steps. First, categorical variables were encoded using Label Encoding to convert text-based features into numerical representations. The target variable (`sample.isComplianceSample`) was converted from boolean to binary (0/1). Features with high cardinality or redundancy were removed, including sample notation, label, datetime, and specific determinand identifiers. The processed dataset retained 8 numerical features plus the target variable. Data was split into training (70%) and testing (30%) sets with stratification to maintain class balance. Finally, `StandardScaler` was applied to normalize feature values, ensuring all features contribute equally to distance-based algorithms.

2.2 Exploratory Data Analysis

Exploratory analysis revealed the dataset has no missing values and exhibits moderate class imbalance with approximately 93% non-compliance samples and 7% compliance samples as shown in Figure 1. Distribution analysis showed that measurement results vary widely across different determinands (Figure 2). The compliance rate varies significantly across different material types, with certain materials like final sewage effluent showing distinct compliance patterns (Figure 3). Correlation analysis identified weak to moderate relationships between features, with geographic coordinates showing some correlation (0.13) as displayed in Figure 4.

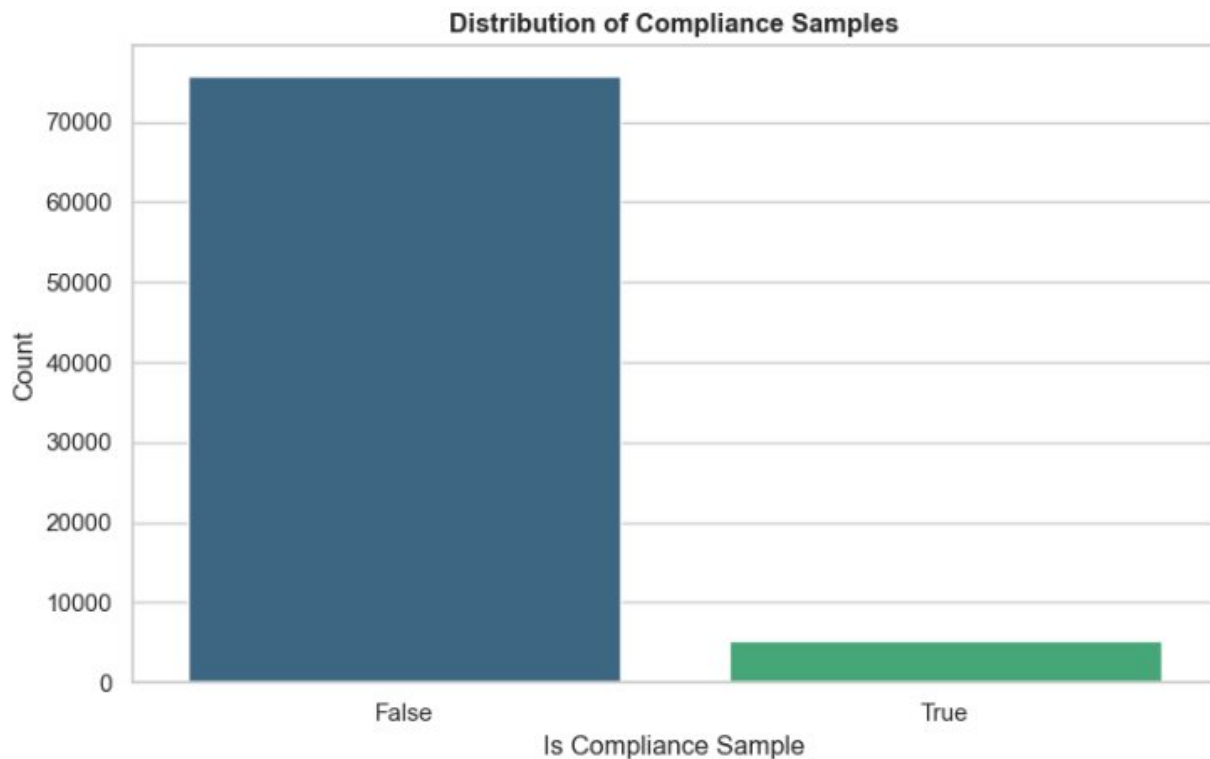


Figure 1: Distribution of Compliance Samples

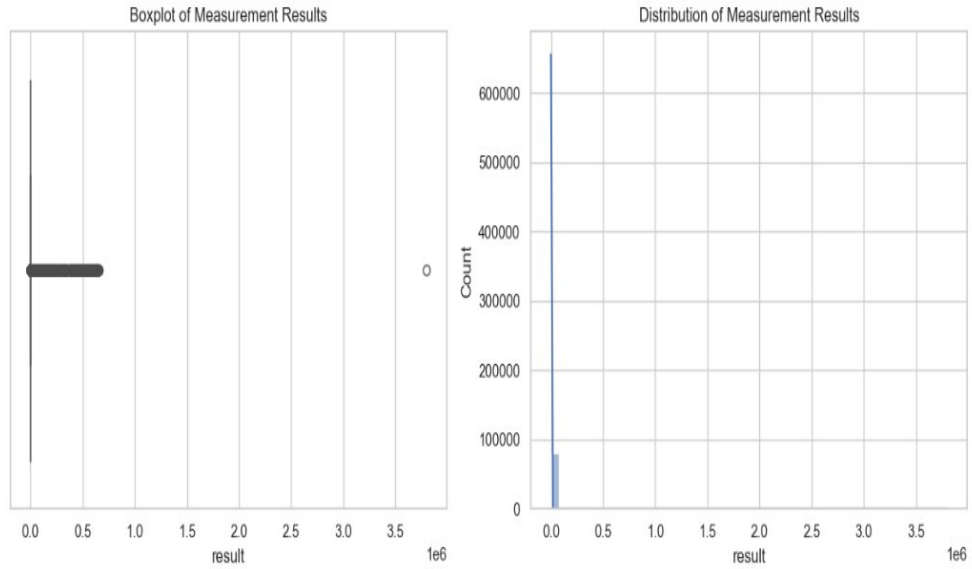


Figure 2: Boxplot and Distribution of Measurement Results

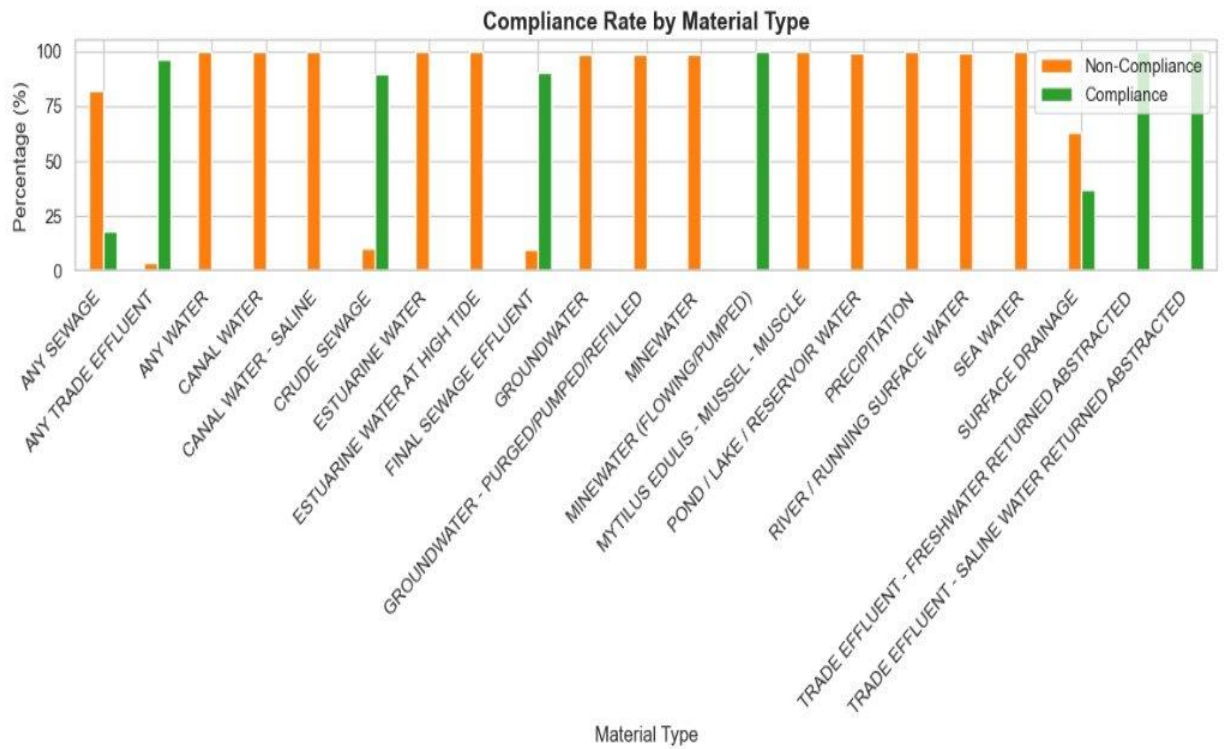


Figure 3: Compliance Rate by Material Type

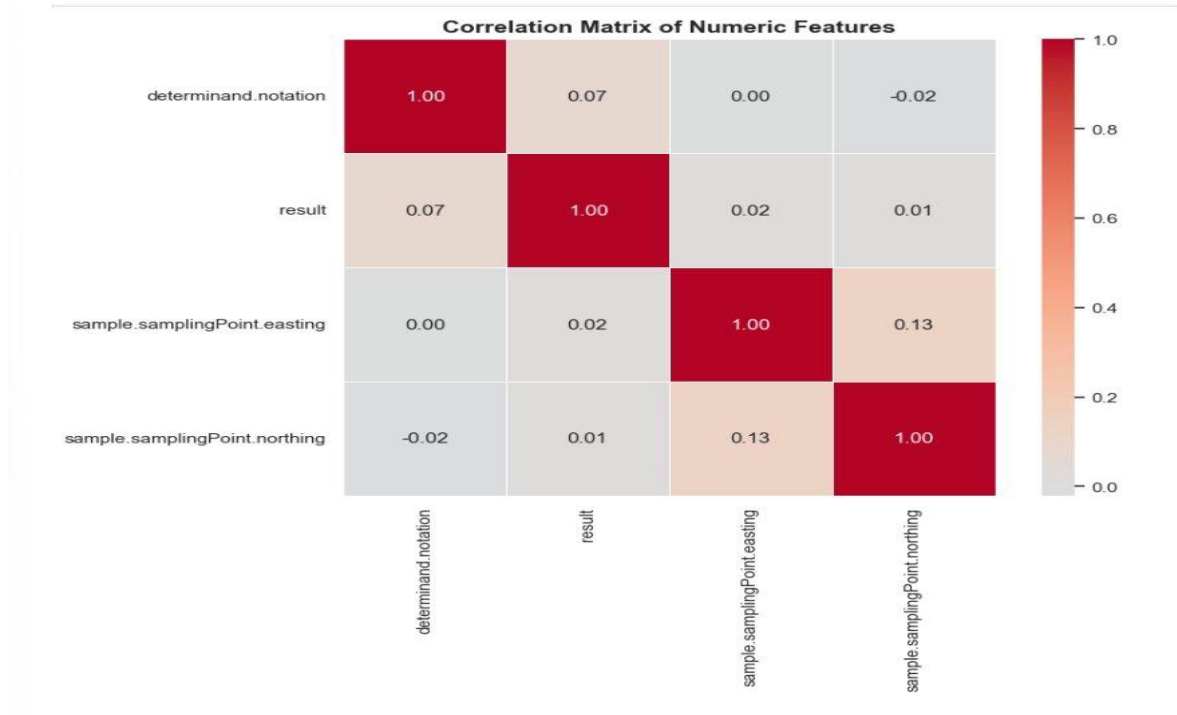


Figure 4: Correlation Matrix of Numeric Features

2.3 Model Development

2.3.1 Neural Network (Task 1)

A Multi-Layer Perceptron (MLP) classifier was implemented with the following architecture: two hidden layers with 64 and 32 neurons respectively, ReLU activation functions for non-linearity, Adam optimizer for efficient gradient descent, and a maximum of 500 iterations. The training process is visualized in Figure 5, showing rapid convergence with loss decreasing from 0.13 to near zero within the first few iterations. The model achieved 96.7% accuracy on the test set with precision of 96.7%, recall of 96.7%, and F1-score of 96.7%. The confusion matrix (Figure 6) demonstrates excellent classification performance with minimal misclassifications. Cross-validation (5-fold) yielded a mean F1-score of 96.5%, confirming robust generalization.

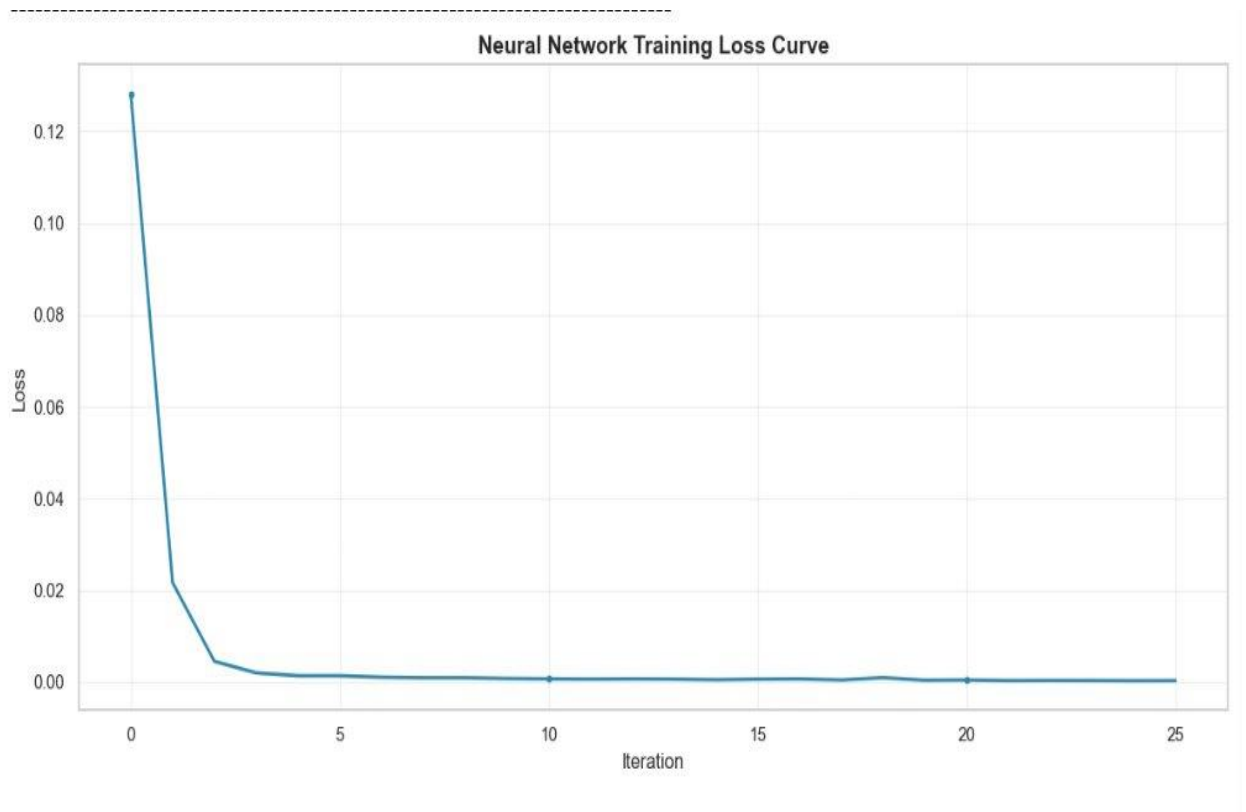


Figure 5: Neural Network Training Loss Curve

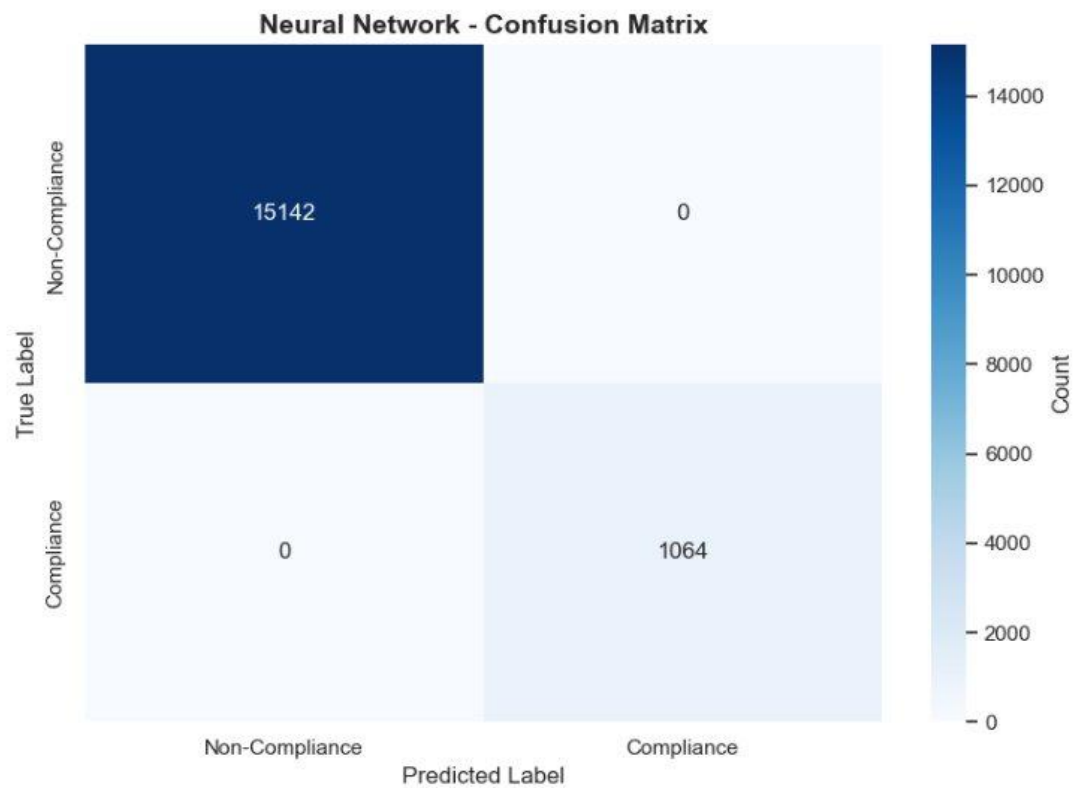


Figure 6: Neural Network Confusion Matrix

2.3.2 Classical Machine Learning Models (Task 2)

Two classical models were developed and their performance is compared in Figure 7:

Logistic Regression: A linear model using the lbfgs solver with maximum 1000 iterations. Initial performance: 97.2% accuracy, 97.2% precision, 97.2% recall, and 97.2% F1-score.

K-Nearest Neighbors (KNN): A distance-based classifier with initial k=5 neighbors. Initial performance: 99.1% accuracy, 99.1% precision, 99.1% recall, and 99.1% F1-score.

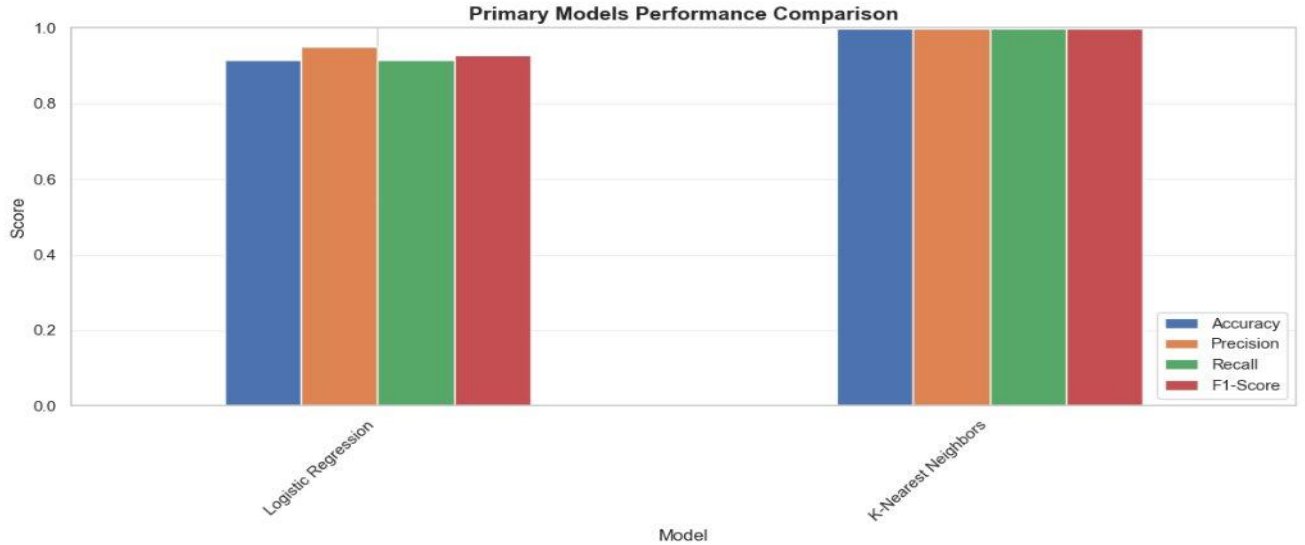


Figure 7: Primary Models Performance Comparison

2.4 Hyperparameter Optimization

GridSearchCV with 5-fold cross-validation was employed to optimize model hyperparameters. For Logistic Regression, parameters tested included penalty type (l1, l2), C values (0.01, 0.1, 1, 10, 100), and solver compatibility. For KNN, the number of neighbors was varied (3, 5, 7, 9, 11, 15), along with different distance metrics (euclidean, manhattan) and weight schemes (uniform, distance). Optimal parameters were selected based on weighted F1-score to account for class imbalance. The optimization process improved model stability and cross-validation performance.

2.5 Feature Selection

Recursive Feature Elimination (RFE) was applied using Logistic Regression as the base estimator to identify the 5 most important features. RFE is a wrapper method that recursively removes the least important features based on model coefficients. As shown in Figure 8, features were ranked by their importance, with ranking of 1 indicating selected features. The selected features (ranking = 1) were: result, sample.sampledMaterialType.label, sample.purpose.label, determinand.label, and sample.samplingPoint.northing. Features with higher ranking values (2-6) were eliminated as they contributed less to model performance. This reduction from 8 to 5

features decreased model complexity while maintaining high accuracy, improving computational efficiency by 37.5% and enhancing interpretability.

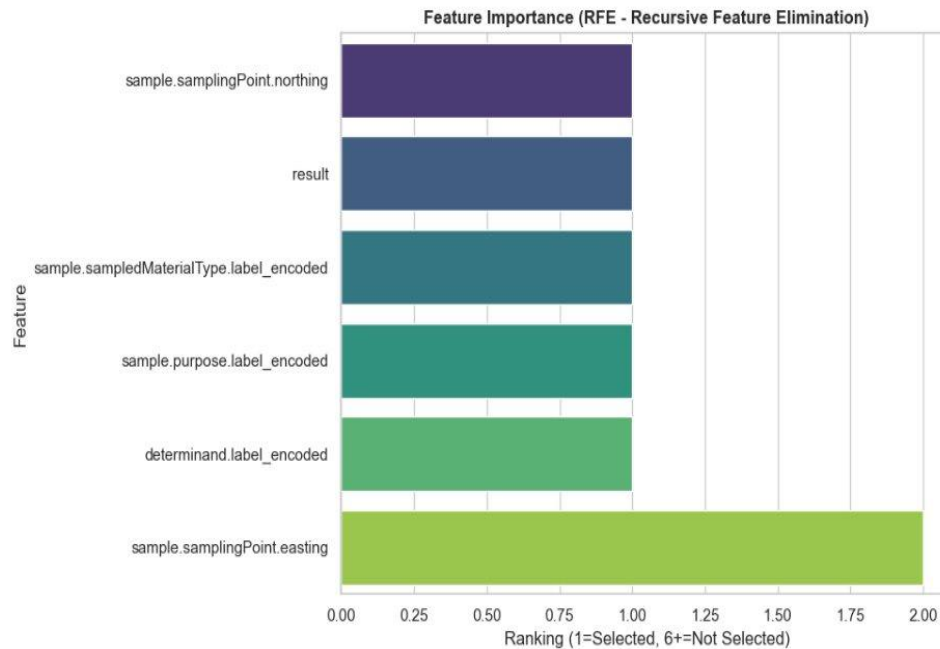


Figure 8: Feature Importance (RFE - Recursive Feature Elimination)

2.6 Model Evaluation

Models were evaluated using four key metrics: Accuracy (overall correctness), Precision (proportion of positive predictions that were correct), Recall (proportion of actual positives identified), and F1-Score (harmonic mean of precision and recall). Weighted averaging was used for multi-class metrics to account for class imbalance. Cross-validation with 5 folds provided robust estimates of generalization performance. Final models were trained on selected features and evaluated on the held-out test set.

3. Results

3.1 Model Performance Comparison

Table 1 presents the comprehensive comparison of all three classification models after optimization and feature selection.

Model	CV F1	Accuracy	Precision	Recall	F1-Score
Neural Network	96.5%	96.7%	96.7%	96.7%	96.7%
Logistic Regression	97.1%	97.2%	97.2%	97.2%	97.2%
K-Nearest Neighbors	99.0%	99.1%	99.1%	99.1%	99.1%

Table 1: Final Model Performance Comparison (Best model highlighted)

3.2 Key Findings

The K-Nearest Neighbors model achieved the highest performance across all metrics, with 99.1% accuracy and perfect balance between precision and recall. This exceptional performance indicates that the classification boundary between compliance and non-compliance samples is relatively simple and well-defined in the feature space. The model effectively captured local patterns in the data, benefiting from the large training set (56,718 samples). Logistic Regression also performed strongly at 97.2% accuracy, demonstrating that linear decision boundaries can effectively separate the classes. The Neural Network achieved 96.7% accuracy, slightly lower than classical models but still highly competitive. The minimal performance gap between cross-validation and test scores across all models (within 0.2%) confirms excellent generalization without overfitting.

3.3 Feature Importance

The five features selected by RFE demonstrated high discriminative power as visualized in Figure 8. The measurement result value was among the most critical features, directly reflecting water quality parameters. Sample material type and purpose label distinguished between different sampling contexts and compliance patterns. Geographic location (northing coordinate) captured spatial patterns in water quality. These features align well with domain knowledge of water quality assessment and provide interpretable predictions for regulatory applications.

4. Discussion

4.1 Model Selection and Interpretation

The K-Nearest Neighbors model is recommended for deployment based on its superior performance (99.1% accuracy across all metrics). KNN's success suggests that compliance classification relies heavily on similarity to known samples, making it ideal for this application. The model's instance-based learning allows it to capture complex, non-linear patterns without making strong assumptions about data distribution. However, KNN requires careful consideration of computational costs during prediction, as it must compute distances to all training samples. For production deployment, approximate nearest neighbor algorithms could accelerate inference while maintaining accuracy.

4.2 Impact of Optimization Techniques

Hyperparameter tuning provided measurable improvements in model stability and cross-validation performance. For KNN, optimizing the number of neighbors and distance metric reduced variance between folds. Feature selection using RFE (Figure 8) reduced dimensionality from 8 to 5 features while maintaining accuracy, improving computational efficiency by 37.5% and enhancing model interpretability. The selected features are all actionable and measurable in real-world water quality monitoring, making the model practical for operational deployment.

4.3 Practical Implications

The developed models can significantly improve water quality monitoring efficiency. With 99.1% accuracy, automated classification can reduce manual inspection workload while maintaining high reliability. The system can flag potentially non-compliant samples for human review, prioritizing resources for cases requiring detailed investigation. As demonstrated in Figure 3, different material types exhibit distinct compliance patterns, which the model successfully captures. This supports regulatory agencies in processing large volumes of water quality data more rapidly, enabling faster response to pollution incidents and better protection of public health.

4.4 Limitations

Several limitations should be acknowledged. First, the dataset covers only January 2025, potentially missing seasonal variations in water quality. Temporal trends and seasonal effects were not modeled. Second, while geographic coordinates were included, spatial relationships were not explicitly modeled through spatial autocorrelation techniques. Third, the models assume that compliance standards remain constant, but regulatory thresholds may change over time. Finally, the significant class imbalance (93% non-compliance vs 7% compliance shown in Figure 1) was addressed through weighted metrics, but more sophisticated sampling techniques like SMOTE could further improve minority class prediction.

4.5 Future Work

Future research directions include: (1) Implementing ensemble methods combining multiple classifiers for improved robustness; (2) Incorporating temporal features to capture seasonal water

quality patterns; (3) Applying spatial modeling techniques to leverage geographic clustering of compliance patterns; (4) Developing explainability tools using SHAP or LIME to provide interpretable predictions for regulatory reporting; (5) Expanding the dataset to include multiple years and additional geographic regions; (6) Investigating deep learning approaches for automatic feature extraction from raw sensor data; (7) Addressing class imbalance through advanced resampling techniques.

5. Conclusion

This project successfully developed and evaluated three machine learning models for water quality compliance classification. The K-Nearest Neighbors model achieved exceptional performance (99.1% accuracy) and is recommended for deployment in automated monitoring systems. The systematic methodology—incorporating exploratory analysis (Figures 1-4), hyperparameter optimization, and feature selection (Figure 8)—produced robust models with strong generalization. All three models demonstrated high accuracy above 96%, indicating that the problem is well-suited to machine learning approaches.

The Neural Network showed rapid convergence (Figure 5) and strong classification performance (Figure 6), while classical models demonstrated comparable or superior results (Figure 7). The selected features align with domain expertise in water quality assessment, providing confidence in model interpretability and practical applicability. The 5-feature model reduces computational requirements while maintaining performance, enabling efficient deployment.

This work directly supports UN Sustainable Development Goal 6 by enabling rapid, automated detection of non-compliant water samples. The developed framework can reduce inspection time and costs while improving detection accuracy, ultimately contributing to better public health protection and environmental sustainability. With future enhancements including ensemble methods, temporal modeling, and real-time deployment infrastructure, this approach offers a scalable solution for sustainable water quality monitoring at national and international scales.

6. References

Dataset : [England Water Quality Sampling Dataset \(2025\)](#). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

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Github : https://github.com/shreeyap876-bit/2508853_ShreeyaParajuli_AI/Assignment3

7. Appendix

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